Technical training.
Product information.

G12 LCI Complete Vehicle

Edited for the U.S. market by:
BMW Group University
Technical Training
ST1901 4/1/2019

BMW Service

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:

⚠

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status: January 2019

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

The information contained in the training course materials is solely intended for participants in this training course conducted by BMW Group Technical Training Centers, or BMW Group Contract Training Facilities.

This training manual or any attached publication is not intended to be a complete and all inclusive source for repair and maintenance data. It is only part of a training information system designed to assure that uniform procedures and information are presented to all participants.

For changes/additions to the technical data, repair procedures, please refer to the current information issued by BMW of North America, LLC, Technical Service Department.

This information is available by accessing TIS at www.bmwcenternet.com.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application
- Aftersales Information Research (AIR)

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G12 LCI Complete Vehicle

1. Introduction

From spring 2019 the G12 are offered within the framework of the Life Cycle Impulse (LCI).

There are new visual features primarily in the outer area. They give the LCI more presence in comparison to its predecessor. Changes to the usual luxurious passenger compartment can only be found in the details.

The technological highlights of the Life Cycle Impulse are the changeover of the user interface to ID7 (7th generation BMW iDrive), the extension of the driver assistance systems to semi-autonomous driving .

BMW G12 LCI

1.1. Models

The following table provides an overview of the available model ranges:

Model	G12 LCI
BMW 740i	X
BMW 740i xDrive	X
BMW 750i xDrive	X
BMW 760i xDrive	X

1.2. Dimensions

The outer dimensions of the G12 LCI are listed below:

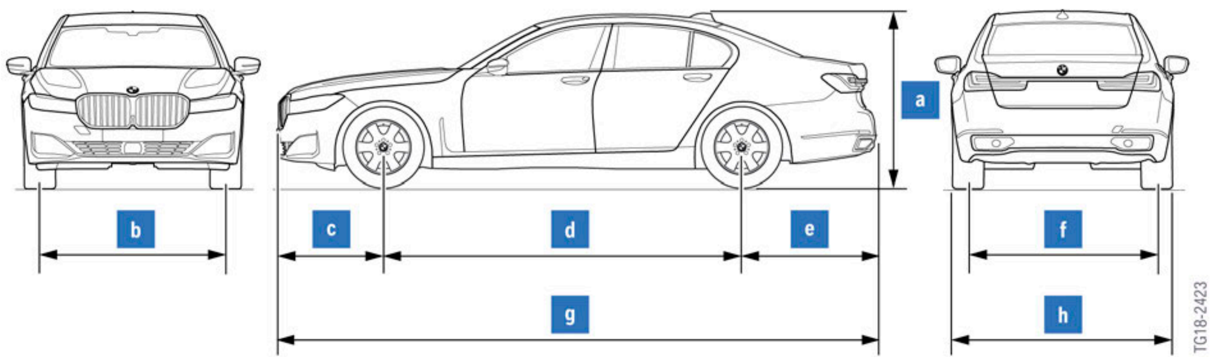


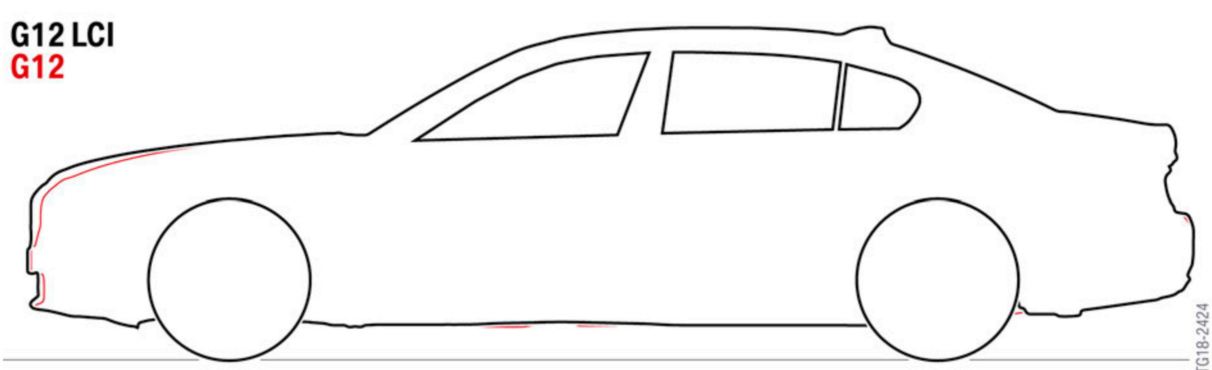
Diagram of G12 LCI

Index	Explanation	G12 LCI [mm]
a	Vehicle height, empty	1479
b	Front track width, basic wheels	1612
c	Front overhang	**909**
d	Wheelbase	3210
e	Rear overhang	**1148**
f	Rear track width, basic wheels	1640
g	Vehicle length	**5267**
h	Width excluding exterior mirrors	1902

Highlighted: Changes to the predecessor model

1.3. Silhouette comparison

A silhouette comparison with the G12 shows the differences in the dimensions compared to the Life Cycle Impulse.



Silhouette comparison between G12 LCI and G12

G12 LCI Complete Vehicle

2. Body

2.1. Exterior equipment

2.1.1. Front

Change

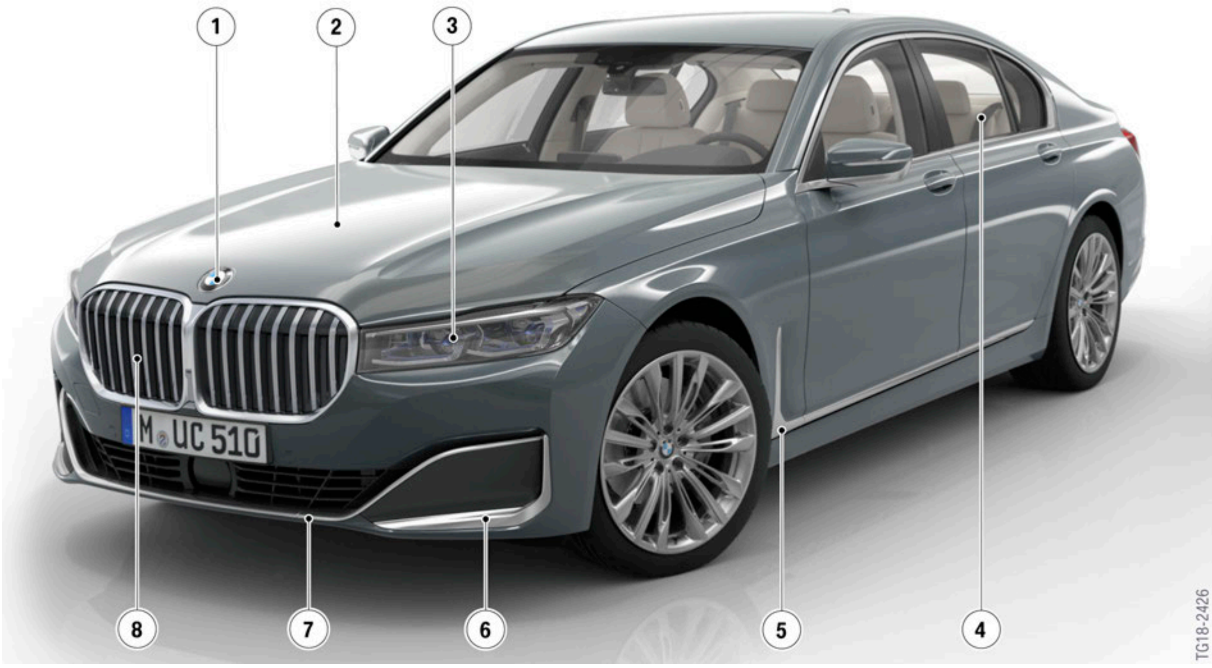
- Larger BMW badge
- Modified radiator grill design
- Design changes to the bumper and the hood
- New headlight assembly design.

Comparison of front of G12 before and after LCI

Index	Explanation
1	G12
2	G12 LCI

G12 LCI Complete Vehicle

2. Body



G12 LCI highlights, exterior equipment

Index	Explanation
1	BMW badge with 95 mm diameter
2	New hood design with more pronounced edges
3	New headlight assembly design
4	Increased material thickness of laminated safety glass on 8- and 12-cylinder engines
5	New air outlet design
6	New trim panel design with concealed air inlet
7	Three-piece trim strip
8	New radiator grill design, significantly enlarged

2.1.2. Rear

Change

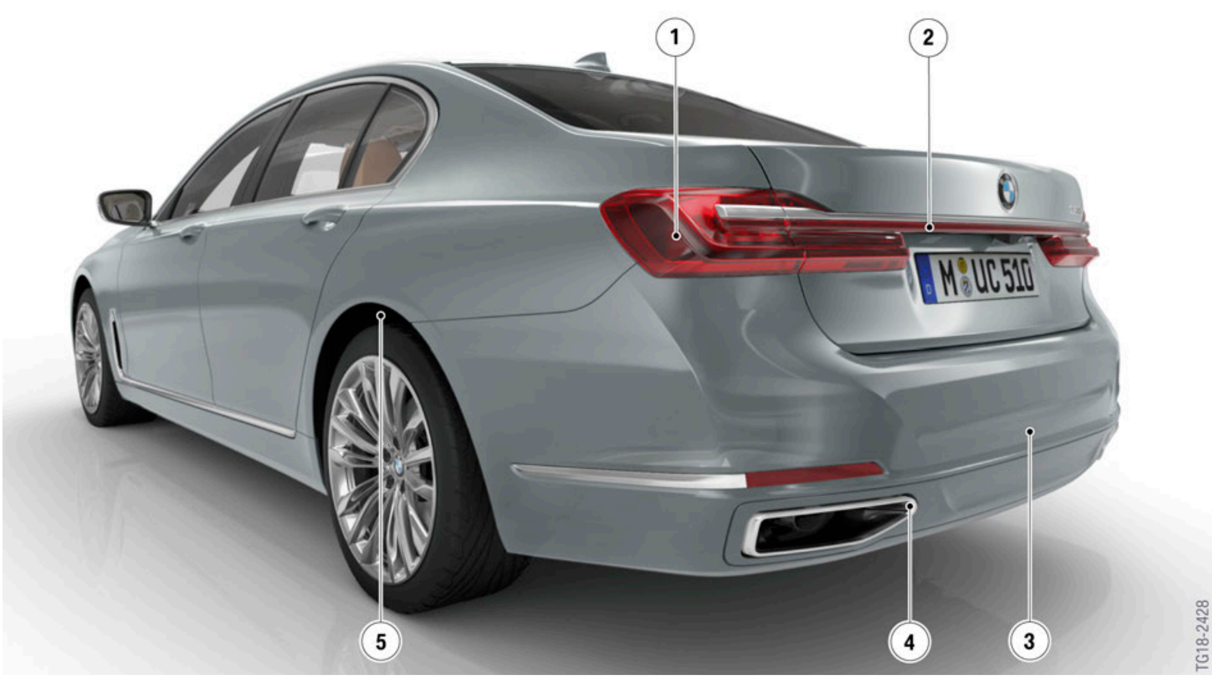
- New bumper design with integrated exhaust pipe trim.
- New rear light design with LED technology.
- Full-length trim strip and rear light cluster at the tailgate.

G12 LCI Complete Vehicle

2. Body

Comparison of rear of G12 before and after LCI

Index	Explanation
1	G12
2	G12 LCI

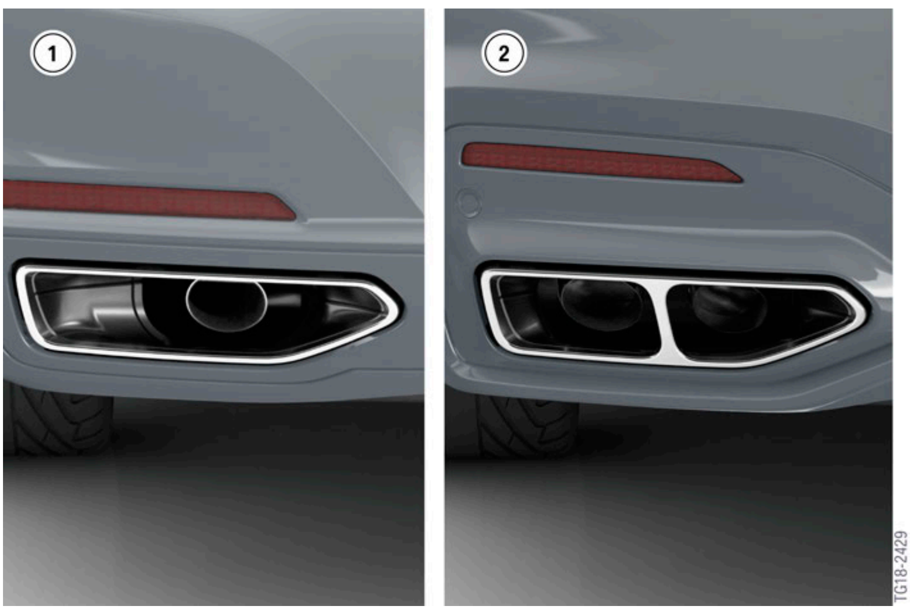


G12 LCI highlights, exterior equipment

G12 LCI Complete Vehicle
2. Body

Index	Explanation
1	New rear light design with LED technology
2	Full-length rear light in the tailgate
3	New bumper design
4	New exhaust pipe trim design
5	New acoustic package for the rear wheel arch

The exhaust pipe trim on the G12 LCI with 12-cylinder engine has been given its own design.



Comparison of the exhaust pipe trim on the G12 LCI

Index	Explanation
1	Exhaust pipe trim for 6- and 8-cylinder engines
2	Exhaust pipe trim for 12-cylinder engine

Two adjustments have been made to the body area for the 750i of the G12 LCI.

The rear diffuser has been adapted to the exhaust system of the 750i.

New rear diffuser adapted to the exhaust system of the 750i with the N63B44T3 engine

The cross-brace has been given a new design in aluminium using pressure die casting technology (design concept taken from the motor sport department).

Cross-brace adapted to the N63B44T3 engine

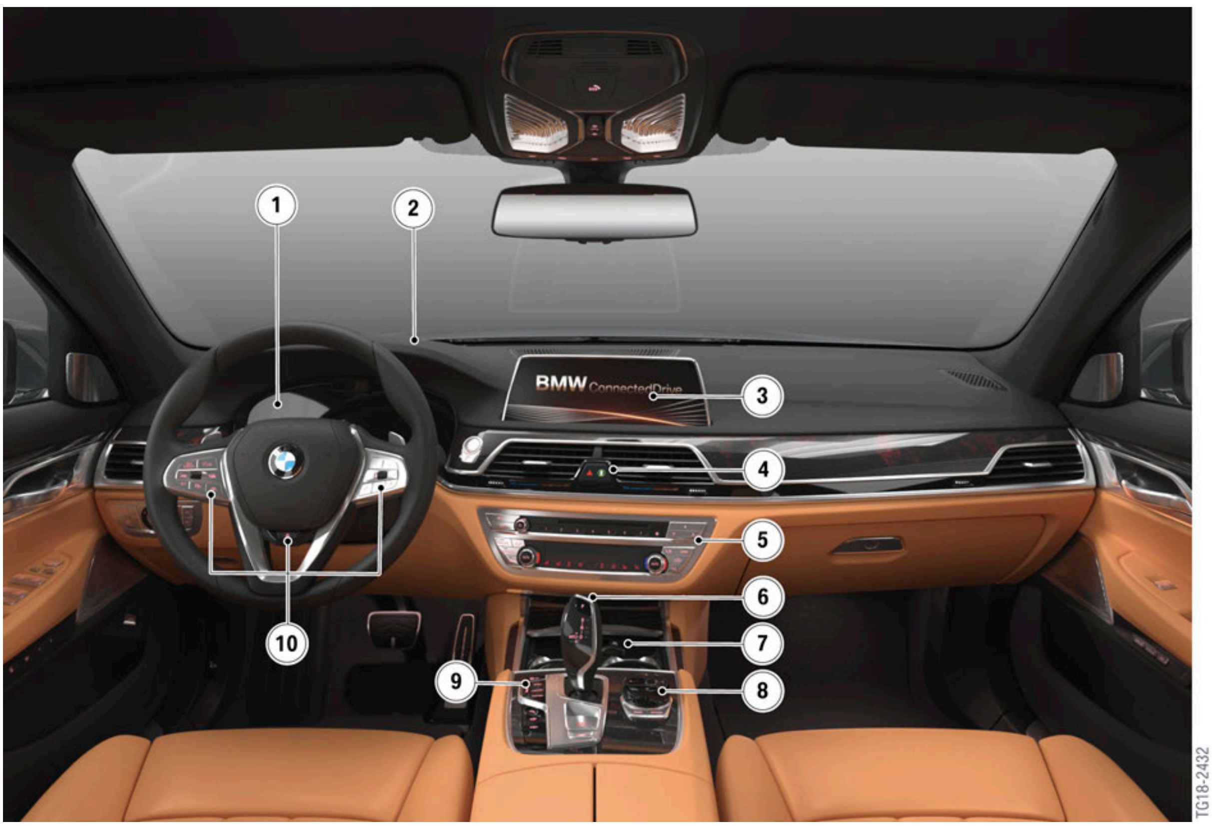
G12 LCI Complete Vehicle
2. Body

2.2. Interior equipment

2.2.1. Overview of the interior

The proven interior design of the G12 has been incorporated almost in its entirety into the LCI.

The only changes can be found at the button layout on the steering wheel and in the front storage compartment on the center console. The wireless charging station is now located here. The steering wheel trim panel is electroplated in all versions of the G12 LCI.



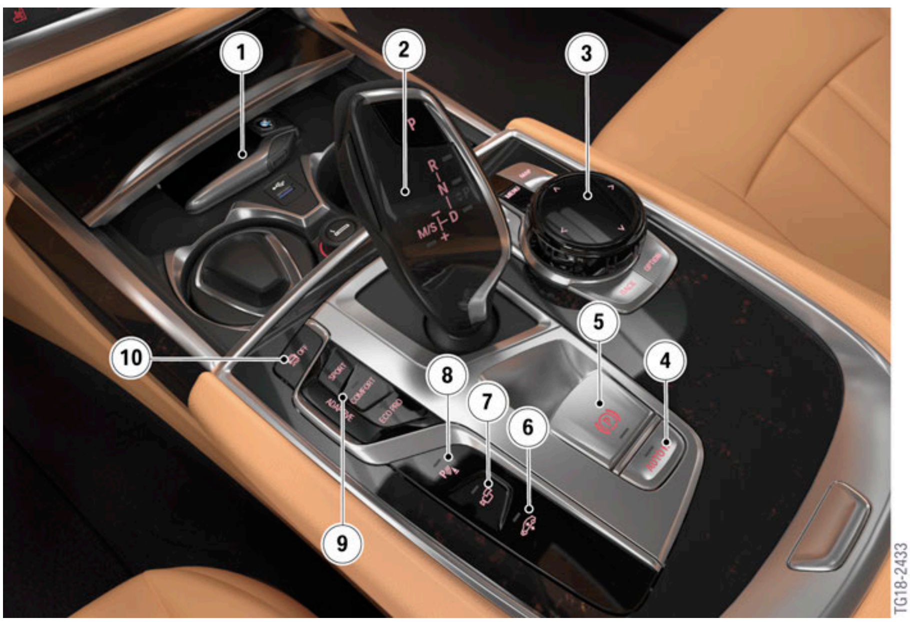
Overview of the passenger compartment of the G12 LCI

Index	Explanation
1	Instrument panel
2	BMW Head-Up Display
3	Central Information Display (CID)
4	Intelligent Safety button
5	Air conditioning / radio controls
6	Gear selector lever

Index	Explanation
7	Wireless charging station
8	Controller
9	Driving experience switch
10	Multifunction steering wheel

2.2.2. Operating elements in center console

The following graphic shows the operating elements in the center console of the G12 LCI:



Overview of center console of the G12 LCI

Index	Explanation
1	Wireless charging station
2	Gear selector lever
3	Controller
4	Automatic Hold button
5	Electromechanical parking brake
6	Two-axle ride level control
7	Surround View
8	Park Distance Control
9	Driving experience switch
10	Dynamic Stability Control

BMW display keys or compatible smartphones can be charged in the wireless charging tray.

2.3. Luggage compartment

With the LCI, the trunk of the G12 has been upgraded by means of a stainless steel trim panel on the loading edge. This basic equipment is not available for the plug-in hybrid electric vehicle.

G12 LCI loading edge with stainless steel trim panel

G12 LCI Complete Vehicle

3. Powertrain

3.1. Engines

The powertrain of the G12 LCI has been fitted with the latest generation of modular engines.

3.1.1. Engines

The G12 LCI will be available with a 6, 8 or 12-cylinder engine.

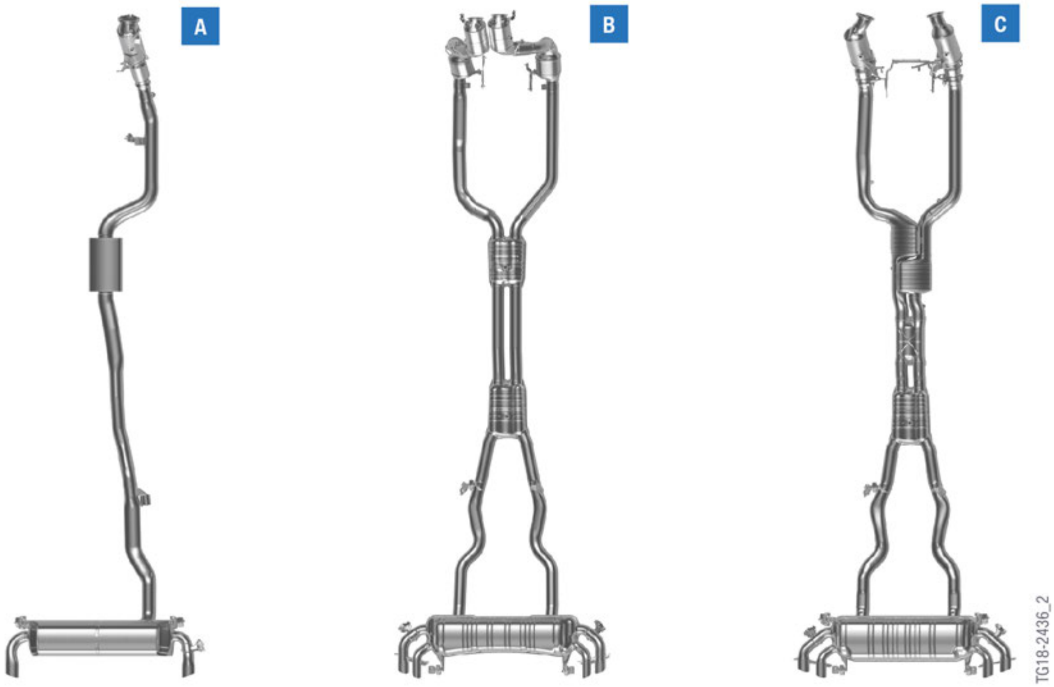
The G12 LCI plug-in hybrid electric vehicle (PHEV) is described in a separate reference manual "ST1902 G12 LCI PHEV" no further reference is made to it here.

The table below presents the technical data of the gasoline engines:

Models	**Engine identification**	**Power output [kW (hp)]**	**Torque [Nm (lb-ft)]**
740i (xDrive)	B58B30M1	**250 (335)**	450 (330)
750i xDrive	N63B44T3	**390 (523)**	**750 (553)**
760i xDrive	N74B66U2	448 (600)	**850 (627)**

Exhaust system

The exhaust systems of the different models are shown below:



Exhaust system variations

Index	Explanation
A	B58B30M1
B	N63B44T3
C	N74B66U2

G12 LCI Complete Vehicle

4. Chassis and Suspension

4.1. Dynamic Stability Control integrated (DSCi)

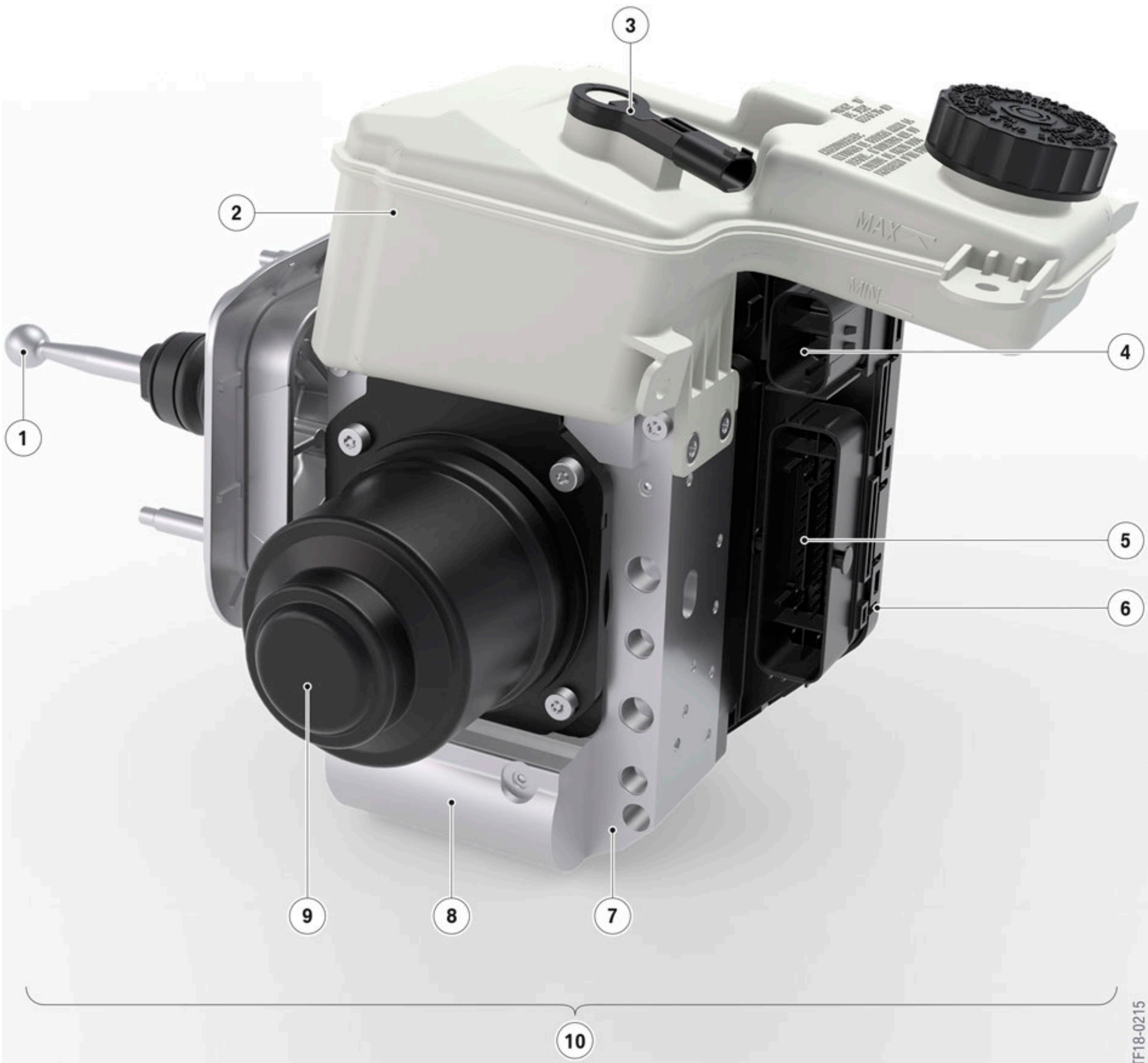
The G12 LCI features the newly developed integrated brake system with the internal designation "Dynamic Stability Control integrated" (DSCi).

With the introduction of the DSCi, a completely new braking concept is available to BMW customers which has a significant effect on driving dynamics.

4.1.1. Features

The brake system is distinguished by the following driving characteristics in particular:

- Outstanding driving dynamics and vehicle control due to the dynamics and precision of the vehicle stabilization.
- More sporting character and feeling of safety due to a brake pedal feel short travel and effective modulation.
- Enhanced active safety thanks to shorter stopping distances in combination with driver assistance systems.
- Due to the fast pressure build-up, much faster and more precise interventions can be achieved compared to previous brake systems.



Dynamic stability control integrated DSCi in the G12 LCI

Index	Explanation
1	Brake control linkage with adjustable ball head
2	Expansion tank
3	Brake fluid level sensor
4	Plug connection, power supply (DC)
5	Plug connection, electrical system
6	Control unit

Index	Explanation
7	Hydraulic unit
8	Brake pedal force simulator
9	3-phase e-motor (AC)
10	DSCi unit

The technical highlight of DSCi is the separation of the driver from the brake hydraulics of the wheel brakes. Brake systems with this attribute are called electro-hydraulic brake-by-wire brakes. Electro-hydraulic brake-by-wire brake systems have been in use at the BMW Group since 2009. However, their range of use was limited to a very small number of models with hybrid drives. With the introduction of DSCi, vehicle models with conventional drive concepts are now also switching to electro-hydraulic brake-by-wire technology.

4.1.2. Functional principle

When the driver presses the brake pedal in brake-by-wire mode, the brake request is sensed via a brake pedal travel sensor. 2 driver separator valves prevent the hydraulic pressure generated from being able to act in the direction of the wheel brake. Instead, the hydraulic pressure passes through the opened simulator valve to the brake pedal force simulator. An elastomer inside the brake pedal force simulator generates the customary counterforce.

The sensor signal from the brake pedal travel sensor is processed by the DSCi control unit and the linear actuator is activated in accordance with the driver's brake request. The generated brake pressure is routed through the opened linear actuator changeover valves in the direction of the brake calipers.

4.1.3. Special features

The new DSCi brake system in the G12 LCI is characterized by the following special technical features:

- Electro-hydraulic brake-by-wire braking function.
- Vacuum supply omitted.
- Vacuum brake servo omitted.
- Integration of the tandem brake master cylinder.
- Integrated brake pedal travel sensor.
- Changeover from front/back to diagonal brake force distribution.
- Changeover from a brake fluid level switch to a brake fluid level sensor.

Further information on DSCi can be found in "ST1852 DSCi" .

4.2. Integrated RDCi tire pressure monitor

The G12 LCI features the familiar tire pressure monitor integrated RDCi which is integrated into the dynamic stability control. There are different system suppliers depending on the vehicle type. This must be taken into consideration when replacing the wheel electronics.

4.2.1. Electronic tire pressures plate

The G12 LCI is fitted with the electronic tire pressure label introduced in the G30 (BMW 5 Series). For the valid tire inflation pressures, check the Central Information Display (CID).

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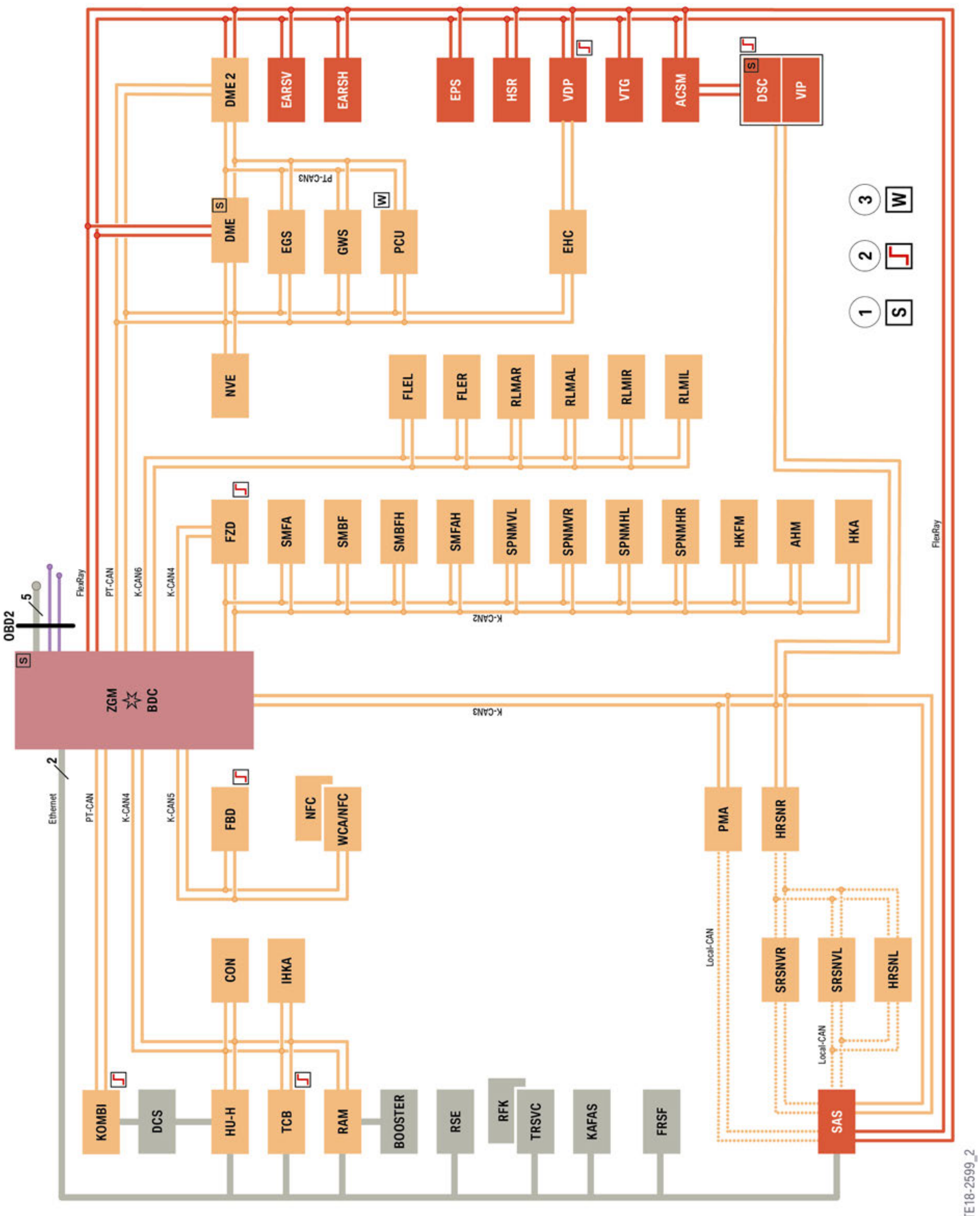
The RDC reset is omitted following adjustment of the tire inflation pressures in vehicles with activated electronic tire pressure specification.

Further information can be found in the "ST1604 G30 Complete Vehicle".

5.1. Bus overview



Bus overview G12 LCI

G12 LCI Complete Vehicle
5. General Vehicle Electronics

Index	Explanation
ACSM	Advanced Crash Safety Module
AHM	Trailer module
BDC	Body Domain Controller
Booster	Booster
CON	Controller
DCS	Driver Camera System
DME	Digital Motor Electronics
DME2	Digital Engine Electronics 2
DSC	Dynamic Stability Control
EARSH	Electric active roll stabilization rear
EARSV	Electrical dynamic drive, front
EGS	Electronic transmission control
EHC	Electronic ride height control
EPS	Electromechanical Power Steering
FBD	Remote control receiver
FLEL	Frontal Light Electronics Left
FLER	Frontal Light Electronics Right
FRSF	Front radar sensor long range
FZD	Roof function center
GWS	Gear selector switch
HU-H	Head Unit High
HKA	Automatic rear air-conditioning and heating
HKFM	Tailgate function module
HRSNL	Rear radar sensor short range left
HRSNR	Rear radar sensor short range right
HSR	Rear axle slip angle control
IHKA	Integrated automatic heating / air conditioning
KAFAS	Camera-based driver assistance systems
KOMBI	Instrument panel
NFC	Near Field Communication
NVE	Night Vision Electronics
PCU	Power Control Unit
PMA	Parking Manoeuvring Assistant
RAM	Receiver Audio Module
RFK	Rear view camera

Index	Explanation
RLMAL	Outer rear light module, left
RLMAR	Outer rear light module, right
RLMIL	Inner rear light module, left
RLMIR	Inner rear light module, right
RSE	Rear Seat Entertainment
SAS	Optional equipment system
SMBF	Front passenger seat module
SMFA	Driver's seat module
SMBFH	Seat module, front passenger's side, rear
SMFAH	Seat module, driver's side, rear
SPNMHL	Seat pneumatics module back left
SPNMHR	Seat pneumatics module back right
SPNMVL	Seat pneumatics module front left
SPNMVR	Seat pneumatics module front right
SRSNVL	Side radar sensor short range front left
SRSNVR	Side radar sensor short range front right
TCB	Telematic Communication Box
TRSVC	Top rear side view camera
VDP	Vertical Dynamic Platform
VIP	Virtual Integration Platform
VTG	Transfer box
WCA/NFC	Wireless charging station with control electronics for Near Field Communication
ZGM	Central Gateway Module
1	Start-up node control units for starting and synchronizing the FlexRay bus system
2	Control units authorised to perform wake-up function
3	Control units also connected at terminal 15WUP

5. General Vehicle Electronics

5.2. Diagnostics

An air-flap control with on-board diagnosis is used for the first time on the G12 LCI.

Air flaps with on-board diagnosis on the G12 LCI

The function of the flap must be checked in accordance with the respective legal requirements and regulations of a country.

For the following countries, the function is emission-relevant and the OBD of the air-flap control is performed (as at October 2018):

- US version

Blockages or mechanical damage can be detected with help of the OBD. To this end, a diagnostic is run after each engine start. If the diagnostic detects a fault occurring twice consecutively, the fault is indicated in the vehicle by the emissions warning light.

5.3. Power window regulators

With the model revision, activation of the power window regulator has been revised. By reducing the speed before reaching the upper or lower end position, the process of opening or closing the window completely runs more smoothly. This further increases comfort.

5.4. Exterior lights

5.4.1. Headlight

The G12 LCI only features Adaptive Full LED headlights.

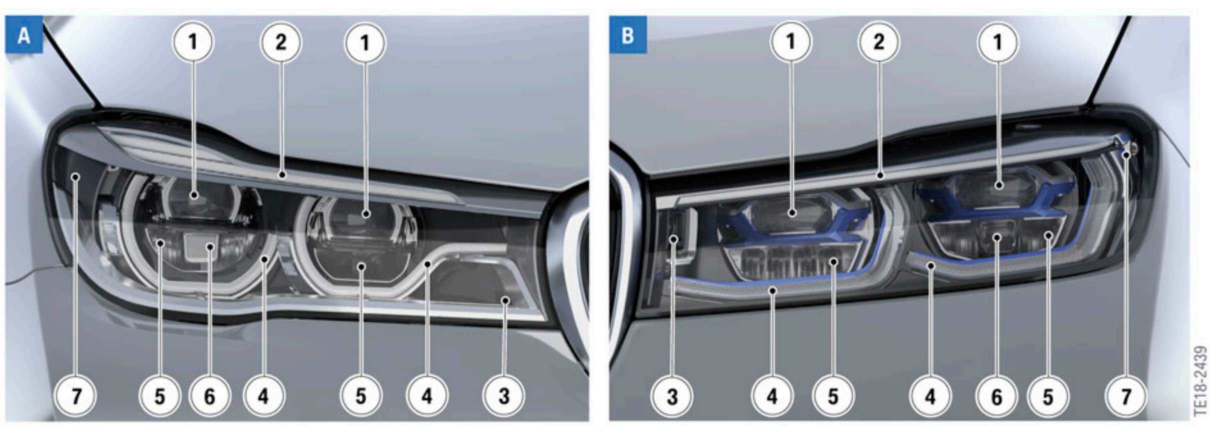
The design has been revised. The turn indicator is now always positioned above the low-beam and high-beam headlight.

Alongside the modified design, the technology remains unchanged. A headlight cleaning system is discontinued.

The following equipment specifications are offered:

- Adaptive Full LED headlights (standard)
- BMW laser light (optional)

Features



Design comparison, using the BMW laser light, between the G12 and G12 LCI

Index	Explanation
A	G12
B	G12 LCI
1	Low-beam headlight
2	Turn indicator
3	Cornering light
4	Side light, daytime driving lights and low-beam headlight
5	High beam
6	Laser light
7	Side marker light

5.4.2. Rear lights

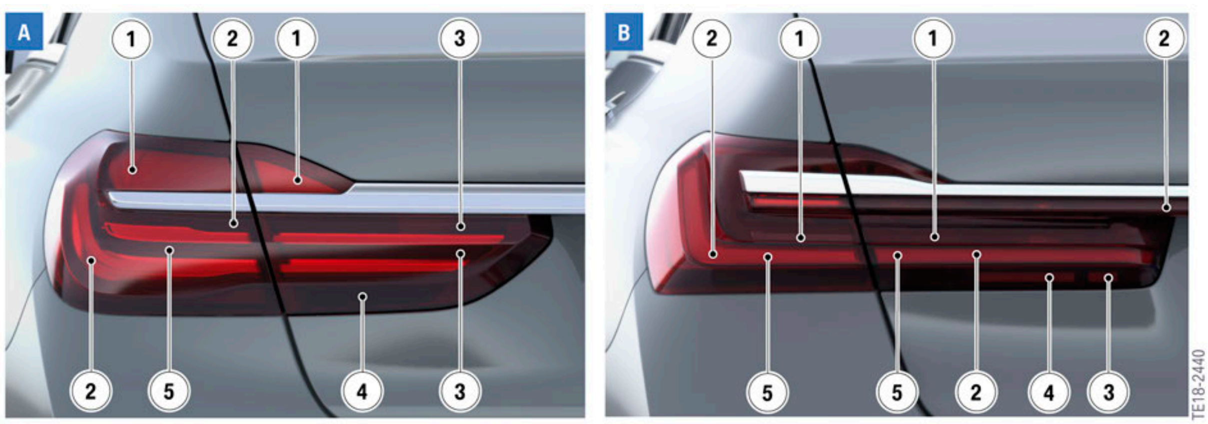
In addition to a new design, the rear lights of the Life Cycle Impulse have full-LED technology.

The rear lights and brake lights are assembled in the same lighting chamber.

The rear light now stretches over the trunk and connects the tail lights on the driver and passenger side.

The turn indicator extends across both light units in the trunk and the side part.

Features



Design comparison of the tail lights between the G12 and G12 LCI

Index	Explanation
A	G12
B	G12 LCI
1	Turn indicator
2	Tail light
3	Rear fog light (Not for the US)
4	Reversing light
5	Brake light

G12 LCI Complete Vehicle
6. Driver Assistance Systems

With the Life Cycle Impulse, the diverse range of driver assistance systems on the G12 is updated and extended even further. The G12 LCI features the vehicle electrical system pack 2018. With the introduction of innovations, we continue along the road to highly-automated driving.

6.1. Further information

This product information presents the new features and modifications to the driver assistance systems in the G12 LCI. The focus is particularly on vehicle-specific features. The following product information items provide basic descriptions of the new features as well as familiar driver assistance systems for specific systems:

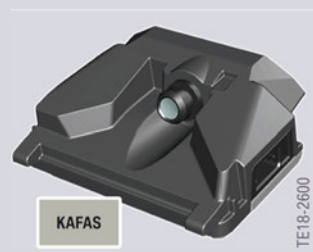
Product information	Information on	
ST1858 Driver Assistance Systems 2018 (new features)	• KAFAS-Mid-Camera • KAFAS-High-Camera • Driver Camera System (DCS) • Collision Warning • Lane Departure Warning • Emergency Stop Assistant	• Junction warning • Active Cruse Control with Stop&Go function • Parking Manoeuvring Assistant PMA • Back-up Assistant • Evasion Aid
ST1831 G05 Driver Assistance Systems (new features)	• Daytime Pedestrian Protection • Junction Warning • Automatic Lane Change • Intersection Collision Warning • Back-up Assistant	• Speed Limit Assistant • Evasion Assistant • Reversing assistant • Emergency Stop Assistant • Lane Departure Warning
ST1604 G30 Driver Assistance Systems (previously published)	• Collision Warning • Speed Limit Info • Intersection Warning • Lane Departure Warning • Active Blind Spot Detection • Cross-traffic Alert front/rear • BMW Night Vision	• Surround View • Remote 3D View • Park Distance Control (PDC) • Parking Manoeuvring Assistant (PMA) • Active Lane Keeping Assistant • Evasion Aid

6.2. Overview

6.2.1. Offer structure "Driving"

The purpose of the following tables is to provide an overview of the dependencies between the offer structure and driver assistance systems used as well as their system components. They also list all driver assistance systems available in the G12 LCI. This overview presents the information status at the production start of the European version of the G12 LCI. Depending on the market and country, other driver assistance systems or function characteristics as well as a different offer structure can be used.

Standard equipment

Active Protection (standard)	
Front Collision Mitigation	

Cruise control with brake function (series)
Manual Speed Limiter

Optional equipment

The following optional equipment is available in the G12 LCI:

- Driving Assistant Professional Package (SA ZDY) includes: Extended Traffic Jam Assistant for Limited Access Highways (SA 5AR) and Active Driving Assistant Pro (SA 5AU).

G12 LCI Complete Vehicle
6. Driver Assistance Systems

Active Driving Assistant Professional
(SA 5AU included in ZDY Driving Assistant Professional Package)

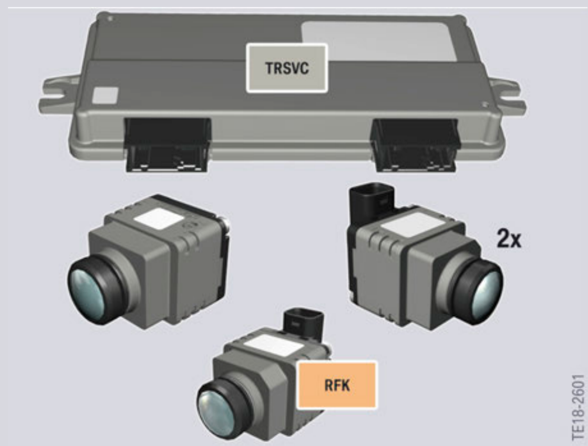
- Lane Departure Warning with Side Collision Mitigation
- Active Cruise Control with Stop&Go
- Automatic Lane Change
- Cross Traffic Warning front
- Evasion Assistant
- Speed Limit Assistant
- Intersection Collision Warning
- Emergency Stop Assistant

Active Driving Assistant (standard)

- Blind Spot Collision Warning
- Cross Traffic Warning Rear
- Speed Limit Info
- Front Collision Mitigation
- Lane Departure Warning
- Daytime Pedestrian Protection

BMW Drive Recorder (SA 6DR)

- Event Recorder
- Crash Recorder

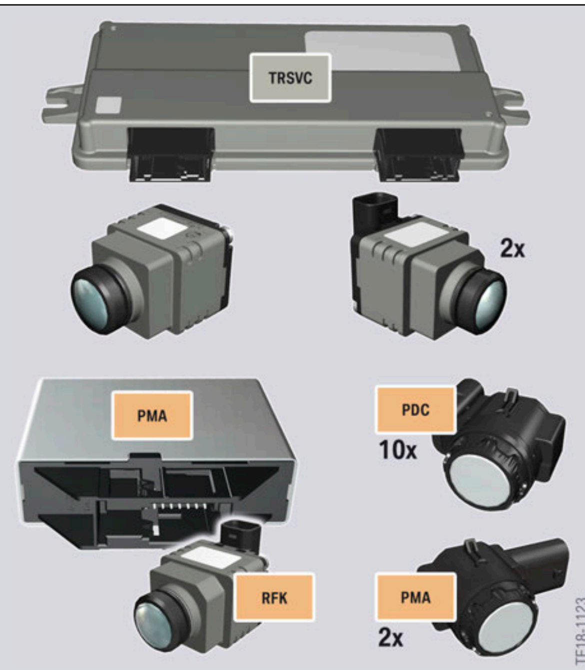


6.2.2. Offer structure "Parking"

The familiar Parking Assistant Plus (SA 5DN) is standard equipment in the new BMW 7 Series after the Life Cycle Impulse.

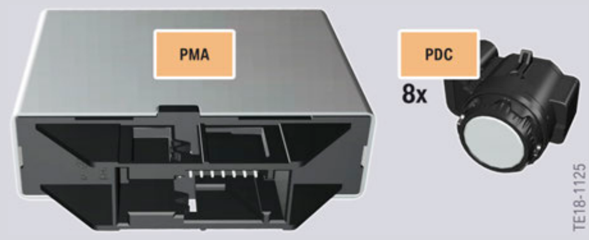
Parking Assistant Plus (standard)

- Surround View
- Panorama View (GPS-based)
- Remote 3D View
- Parking Manoeuvre Assistant (PMA) with lengthwise and crosswise parking and leaving parking space
- Reversing assistant
- Active Park Distance Control (PDC)
- Side protection
- Rear view camera



Active Park Distance Control (PDC) (standard)

- Front and Rear
- Auto PDC



Remote Control Parking (OE 5DV) (not available with the 745e PHEV)

- Remote controlled manoeuvring in and out of a parking space with the BMW display key

6.2.3. Innovations

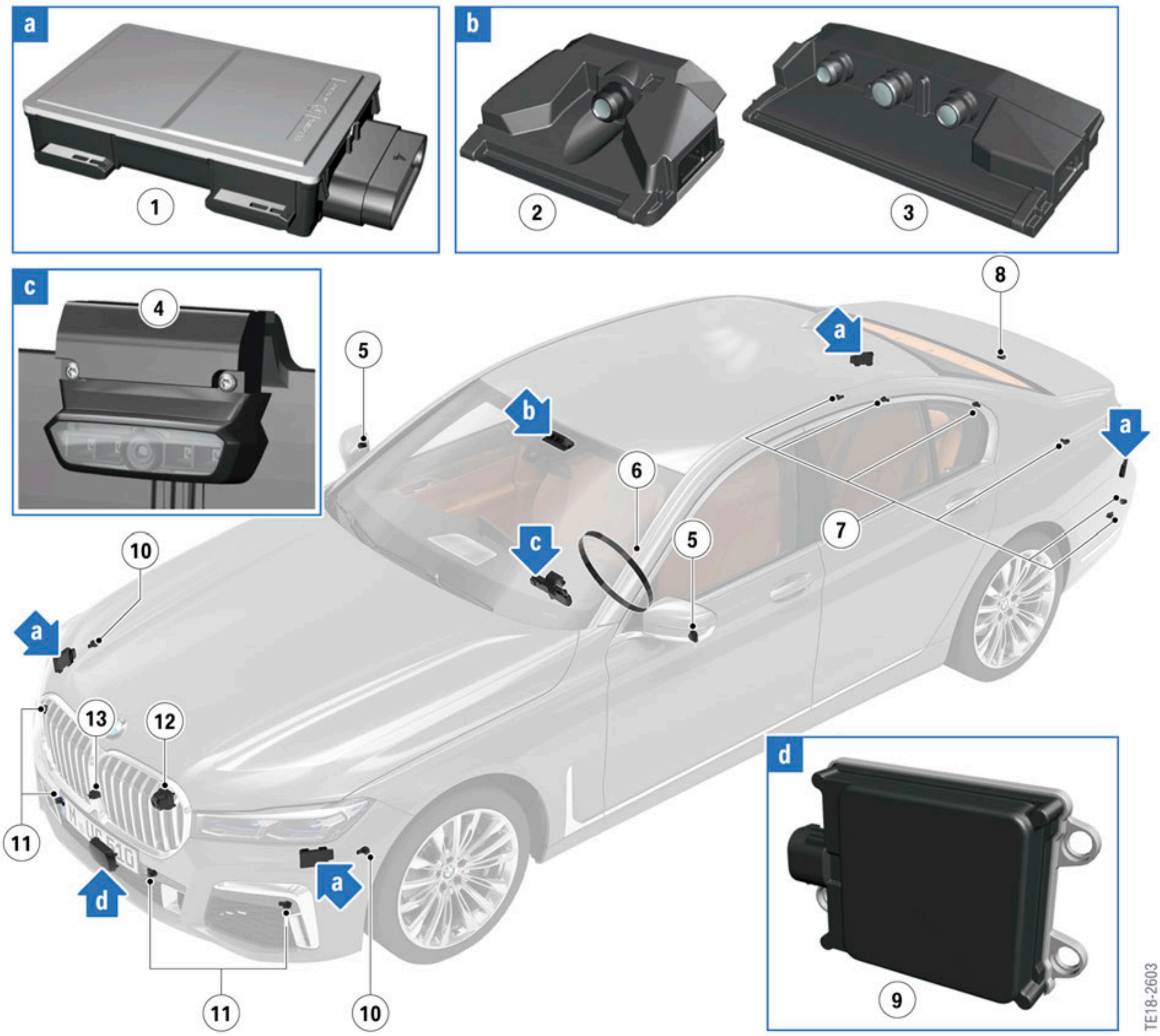
- The instrument cluster has a camera aimed at the driver Driver Camera System (DCS).
- Active Driving Assistant Professional (SA 5AU) included in Driving Assistant professional Package (ZDY).
- A new MODE button in the driver assistance systems control panel on the multifunction steering wheel.
- LED indicators on the steering wheel (only with Driving Assistant Professional SA 5AU).
- With the automatic Speed Limit Assist system, the speed limit ahead can be automatically adopted by the cruise control system.
- The avoidance assistant now also takes pedestrians into account.
- The junction warning has been extended to include a city braking function.
- The emergency stop assistant and lane change assistant are now included in the Driving Assistant Professional (SA 5AU) optional equipment.
- For the first time, the Parking Manoeuvre Assistant (PMA) supports manoeuvring out of parallel parking spaces.
- The parking assistance button no longer has to be held down when parking with the Parking Manoeuvre Assistant.
- The reversing assistant is included in the Parking Assistant Plus.
- The BMW Drive Recorder (SA 6DR) is available as optional equipment.
- The remote-controlled parking function (SA 5DV) can now be activated with the engine running.

G12 LCI Complete Vehicle

6. Driver Assistance Systems

6.2.4. Sensor installation locations

Depending on the vehicle equipment, the sensors shown are used. New or revised sensors are shown enlarged.



G12 LCI overview of driver assistance systems sensors

Index	Explanation
1	Side radar sensors (HRSNR, HRSNL, SRSNVR, SRSNVL)
2	KAFAS-Mid-Camera
3	KAFAS-High-Camera
4	Driver Camera System (DCS)
5	Side view camera
6	Capacitive sensor mat on steering wheel rim
7	Ultrasonic sensors for Park Distance Control (PDC), rear

Index	Explanation
8	Rear view camera (RFK)
9	Front radar sensor long range (FRSF)
10	Ultrasonic sensors for Parking Manoeuvre Assistant (PMA)
11	Ultrasonic sensors for Park Distance Control (PDC), front
12	Night Vision camera
13	Front camera

6.3. Operating elements

When driving, the assistance systems are operated by means of 4 operating elements:

- Light operating unit
- Controls on the multifunction steering wheel
- Intelligent Safety button
- Buttons in the center console

G12 LCI driver assistance systems operating elements

Index	Explanation
1	Light operating unit
2	Control panel for driver assistance systems on the multifunction steering wheel
3	Intelligent Safety button
4	Panorama View button
5	Parking assistance button

The settings within the Intelligent Safety menu are made via the Controller. This chapter only deals with the operating elements relevant to the driver assistance systems. You can find a full description of the display and operating elements of the G12 LCI in the "ST1501 G12 Complete Vehicle / Displays and Controls".

6.3.1. Light operating unit

The Night Vision button for activation of the thermal image screen in the Central Information Display (CID) is located in the light operating unit.

G12 LCI night vision button for thermal image screen in the CID

Index	Explanation
1	Night Vision button

6.3.2. Multifunction steering wheel

G12 LCI operating panel for driver assistance systems on the multifunction steering wheel

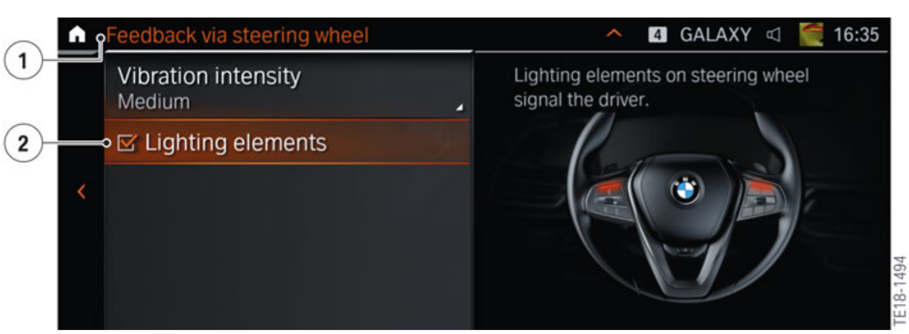
Index	Explanation
A	Operating panel standard equipment (market-specific)
B	Operating panel for "Active cruise control with Stop&Go function" optional equipment
C	Operating panel for "Active Driving Assistant Pro" optional equipment (SA 5AU) includes "Active cruise control with Stop&Go function"

A detailed explanation of the individual buttons and their function can be found in "ST1831 G05 Driver Assistance Systems".

With the Driving Assistant Professional optional equipment (SA 5AU), the number of driver assistance systems exceeds the number of buttons on the multifunction steering wheel. That is why the control of the driver assistance systems has been changed. The assistance system is selected using the MODE button. The selection is then confirmed with the Assist button (to the left).

With the optional equipment Driving Assistant Professional (SA 5AU) there is an LED above the left and right control panel respectively. The two LEDs supplement the displays on the instrument cluster and the instructions on the Central Information Display (CID).

- Green: The assistance system is active and is carrying out the lateral guidance (market-specific)
- Yellow: Interruption of assistance system is imminent
- Red: The assistance system is deactivated.



G12 LCI settings menu LED light elements on the CID

Index	Explanation
1	Menu "Feedback via steering wheel"
2	Light-emitting elements at the multifunction steering wheel (switch on and off)

The LEDs can be deactivated via the iDrive menu:

- "Settings"
- "Driver assistance"
- "Feedback via steering wheel"
  "Light-emitting elements"

6.3.3. Intelligent Safety Button

The Intelligent Safety button, already known from the G12, has been adopted without change. It facilitates central operation of the driver assistance systems. The systems can be switched on or off directly and the Intelligent Safety menu can also be called to personalize the settings via the Intelligent Safety button.

6.3.4. Parking Assistance button

The parking assistance button no longer has to be pressed and held while manoeuvring into a parking space with the Parking Manoeuvring Assistant (PMA). It is sufficient to activate this once.

6.4. Emergency Stop Assistant

The Emergency Stop Assistant is a component of the optional equipment Driving Assistant Professional (SA 5AU) and is used for the first time with the vehicle electrical system Service Pack 2018.

A detailed description of the system can be found in the "ST1858 Driver Assistance Systems 2018".

6.5. Automatic Lane Change

The automatic lane change was introduced for the first time with the G05 and is now part of the Driving Assistant Professional optional equipment (SA 5AU).

A detailed description of the system can be found in the "ST1858 Driver Assistance Systems 2018".

6.6. Speed Limit Assistant

With the introduction of the vehicle electrical system service pack 2018 in the G12 LCI, the Speed Limit Assistant has become a part of the Active Driving Assistant Pro optional equipment (SA 5AU). Additional information about the Speed Limit assist can be found in "ST1831 G05 Driver Assistance Systems".

6.7. BMW Drive Recorder

The BMW Drive Recorder (SA 6DR) is available for the first time on the G12 LCI. This function videos to be recorded around the vehicle and played back on the Central Information Display (CID). Important data, such as vehicle speed, are stored synchronously to the video.

The following applications are possible with the function:

- Manual/event recording of 10 videos with a length of up to 40 seconds
- Automatic recording in the event of an accident while driving. [1]

[1]The function can be activated or deactivated via the controller.

Serious accidents resulting in damage to the hard disk or the cameras can cause the function to be limited or even to fail.

6.7.1. Functional principle

The BMW Drive Recorder optional equipment uses 4 cameras at the front, in the exterior mirrors and at the rear to generate 360° videos of the area around the vehicle. The videos are processed in the TRSVC control unit, stored in the headunit and can be played back on the CID or stored on a USB stick. Important driving data, such as speed and GPS position, are stored along side the video.

The cameras at the front, on the side mirrors and at the rear record continuously. The video is stored on the hard disk of the head unit. The monitoring is continuously overwritten after a 20 second interval.

A permanent recording can then be activated by the customer or is automatically triggered by a crash message from the sensors. In this case, the 20 seconds before activation is retained.

6.8. Remote Control Parking

In the G12 LCI, the remote control parking has been developed further (SA 5DV). The basic operation has remained the same. Driving readiness need no longer be switched off with the start / stop button in order to activate the function. Control of the vehicle can also be taken over by the driver with the engine running following a remote manoeuvre to exit a parking space.

G12 LCI Complete Vehicle
7. Infotainment

With the Life Cycle Impulse, the G12 features the vehicle electrical system service pack 2018. Unlike the new BMW X5 with model designation G05 for example, the G12 LCI features the Head Unit High 3 (HU-H3) with integrated DVD drive.

The performance capacity therefore corresponds to the current status of the technology. The highlights include the processor, RAM and the disk capacity. Thanks to a processor with 2.4 GHz and 4 cores, the Head Unit High 3 therefore has a 6 GB RAM and a 320 GB hard disk.

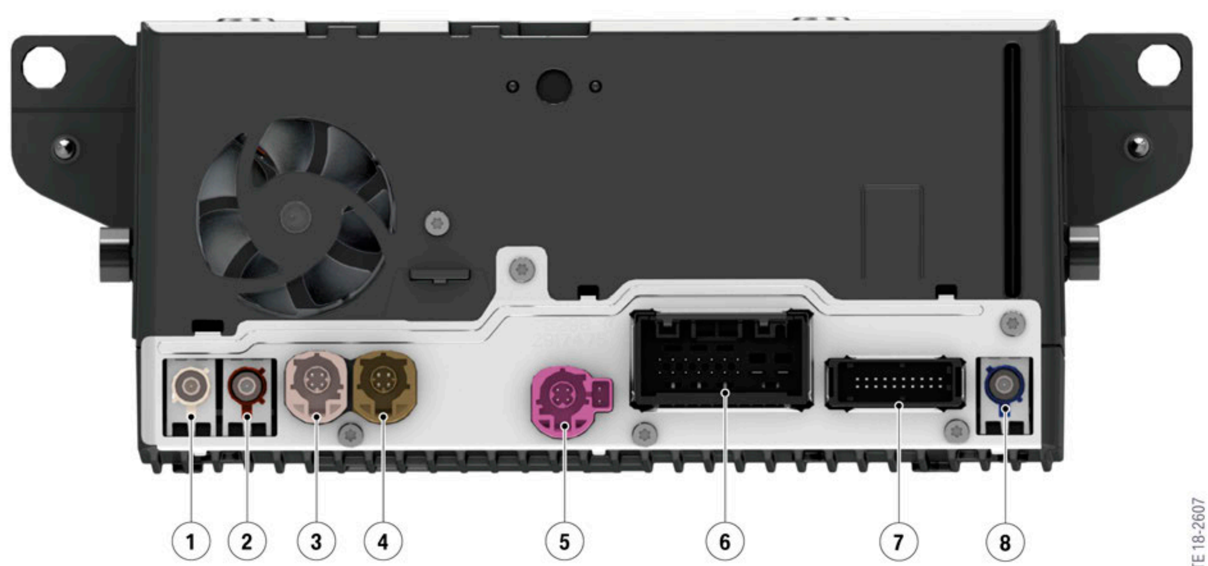
To offer the customer access to technical developments as quickly as possible, the components and control units have been restructured and given a modular design. As with the G05, the G12 LCI also has RAM via the external Receiver Audio Module.

7.1. Head Unit High 3 (HU-H3)

7.1.1. Hardware

In the G12 LCI, the Head Unit High 3 is still housed in the 1.5” DIN housing. The front panel with the DVD drive is unchanged.

Rear view



The Head Unit High 3 rear view in the G12 LCI

Index	Explanation
1	Bluetooth antenna
2	WLAN antenna (vehicle WLAN/Wi-Fi Direct)
3	USB port, type A
4	USB port, type C

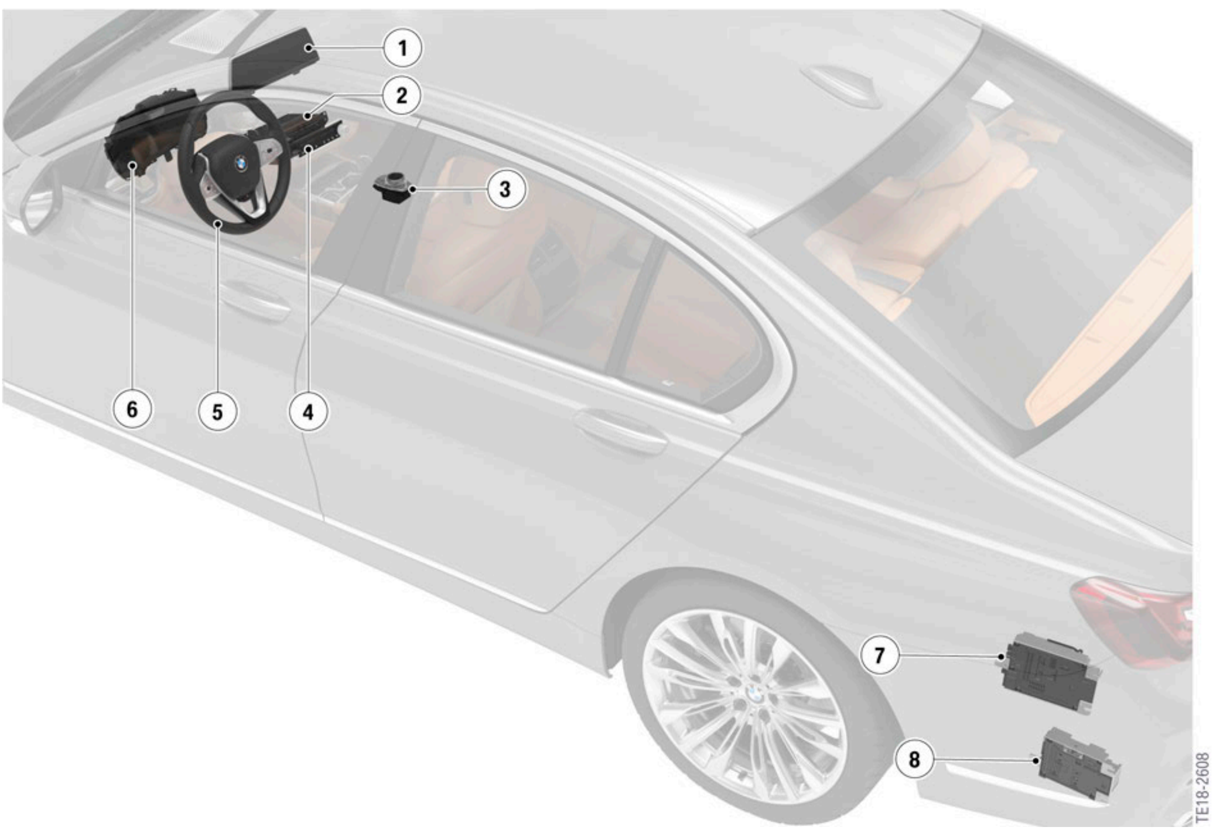
G12 LCI Complete Vehicle
7. Infotainment

Index	Explanation
5	APIX connection to the Central Information Display (CID)
6	Main connector
7	Ethernet connection
8	GPS antenna

Further information on the HU-H3 can be obtained from the "ST1857 Infotainment 2018".

7.1.2. System components

Here you can find an overview of the system components of the head unit high 3.



G12 LCI overview of Head Unit High 3 system components

Index	Explanation
1	Central Information Display (CID)
2	Head unit
3	Controller (CON)
4	Audio operating unit

Index	Explanation
5	Multifunction steering wheel (MFL)
6	Instrument cluster (KOMBI)
7	Receiver Audio Module (RAM)
8	Booster

7.1.3. USB port

USB connections are located in the front center console between the cup holders and the storage compartment under the center armrest.

The connection in center console is a USB type A and delivers 1.5 A charging current.

The connection under the center armrest is USB type C and delivers 3 A charging current.

The data transfer is performed with the standard USB 2.0.

7.2. Receiver Audio Module (RAM)

The receiver-audio module was used for the first time in the G05.

Depending on the equipment, the following functions are integrated in the RAM:

- AM/FM tuner
- SDARS tuner
- Antenna diversity
- Audio amplifier (stereo system, hi-fi system).

Further information on the RAM can be found in "ST1857 Infotainment 2018"

7.3. Booster

The booster is used in the G12 LCI and serves as an additional audio amplifier in the vehicle.

Further information on the booster can be found in "ST1857 Infotainment 2018".

7.4. Navigation system

With regard to the navigation hardware, the navigation system as usual consists of the fixed Central Information Display (CID) with touch function and a resolution of 1920 x 720 pixels, the Head Unit High 3 (HU-H3) and the iDrive controller (CON). With its button assignment, the controller corresponds to the known model from the G12.

For more information, such as new features of the system, can be found in "ST1857 Infotainment 2018".

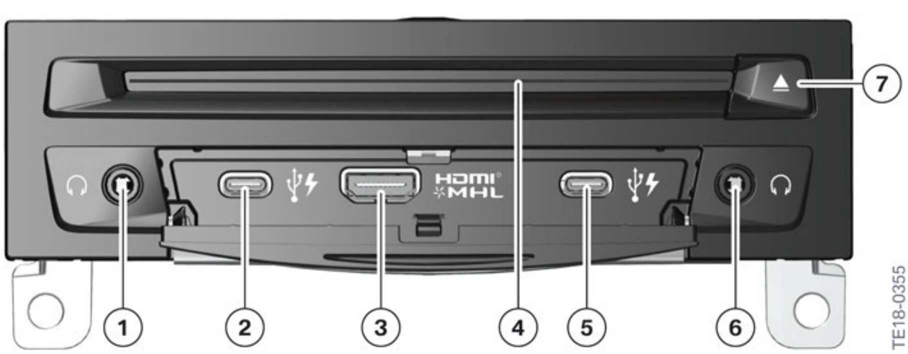
7.5. Rear seat entertainment professional

The concept of the rear seat entertainment professional is retained. With the Life Cycle Impulse, the following changes are incorporated into the rear seat entertainment experience optional equipment (SA 6FR):

- 10.2” monitor with a resolution of 1080p in the backrests of the front seats
- Head Unit High 3 with 2.4 GHz processor and 4 cores
- 8 GB RAM
- 32 GB flash memory
- 2 type C USB ports, each with 3 A charging current and data transfer in accordance with standard USB 2.0
- Resolution of the zone separation
- Access to sources in the front head unit (CD, DVD, USB and Bluetooth audio).

7.5.1. Head Unit High 3

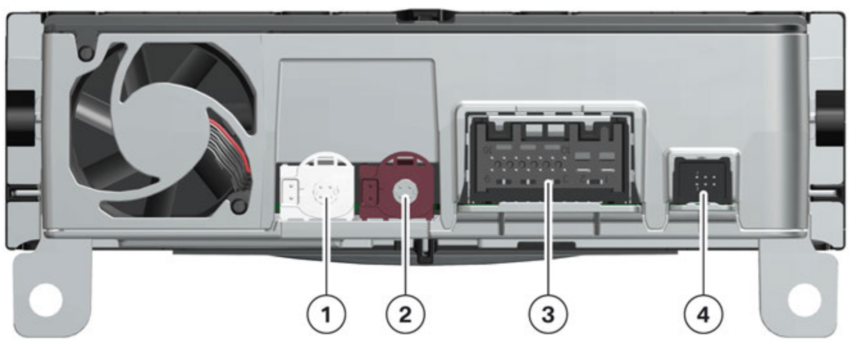
Front view



Front view of RSE HU-H3

Index	Explanation
1	Headphones socket, left
2	USB port, type C, left
3	HDMI/MHL connection
4	Blu-ray drive
5	USB port, type C, right
6	Headphones socket, right
7	Eject button for Blu-ray drive

Rear view



Rear view of RSE HU-H3

Index	Explanation
1	APIX connection for rear compartment display, right
2	APIX connection for rear compartment display, left
3	Main connector
4	Ethernet connection

7.5.2. Touch Command Gen. 2

The 1st generation of the Touch Command was based on the Galaxy Tablet 4 and was manufactured by SAMSUNG up to 12/2016. Also no current software is offered by Android, which can be run on the operating system of the Touch Command Gen. 1 (Android KitKat).

For this reason a new version of the Touch Command is available.

Comparison of technical data

Name	1st generation	2nd generation
CPU	1.2 GHz ARM A7 Quad Core	1.6 GHz Octa Core
Memory	8 GB	8 GB
Working memory RAM	1.5 GB	1.5 GB
Display	7" 1280 x 800 pixels	7" 1280 x 800 pixels (Anti-Reflex-Display)
Android operating system	KitKat 4.4.2	Marshmallow 6.0
Dimensions [mm]	186.9 x 107.9 x 9 (no adapter plate)	186.9 x 107.9 x 12.3 (integrated adapter)
Weight [g]	276 (incl. adapter plate 336)	305
Battery [mAh]	3050	3050

Name	1st generation	2nd generation
Screen mirroring	No	Yes
Temperature range [°C]	-20 to +60	-20 to +60
USB	2.0	2.0

Adapter plate

In the first generation of the Touch Command the table officially marketed by Samsung® was adapted to the needs in the vehicle. For this purpose an adapter plate was attached at the rear side of the tablet. Via the adapter plate it was detected whether the Touch Command was in the snap-in adapter and if it was correctly inserted.

Deletion of adapter plate

The tablet was technically revised for the Touch Command of the 2nd generation. In addition to enhanced performance capabilities, the geometry of the hardware was adjusted as well. The 2nd generation Touch Command fits into the vehicle's snap-in adapter without an adapter plate. The 2nd generation Touch Command is downwardly compatible and replaces the 1st generation Touch Command with an exchange.

7.5.3. Touch Command ID7

The display of the Touch Command was adapted to the display and operating concept ID7.

G12 LCI Complete Vehicle
8. Displays and Controls

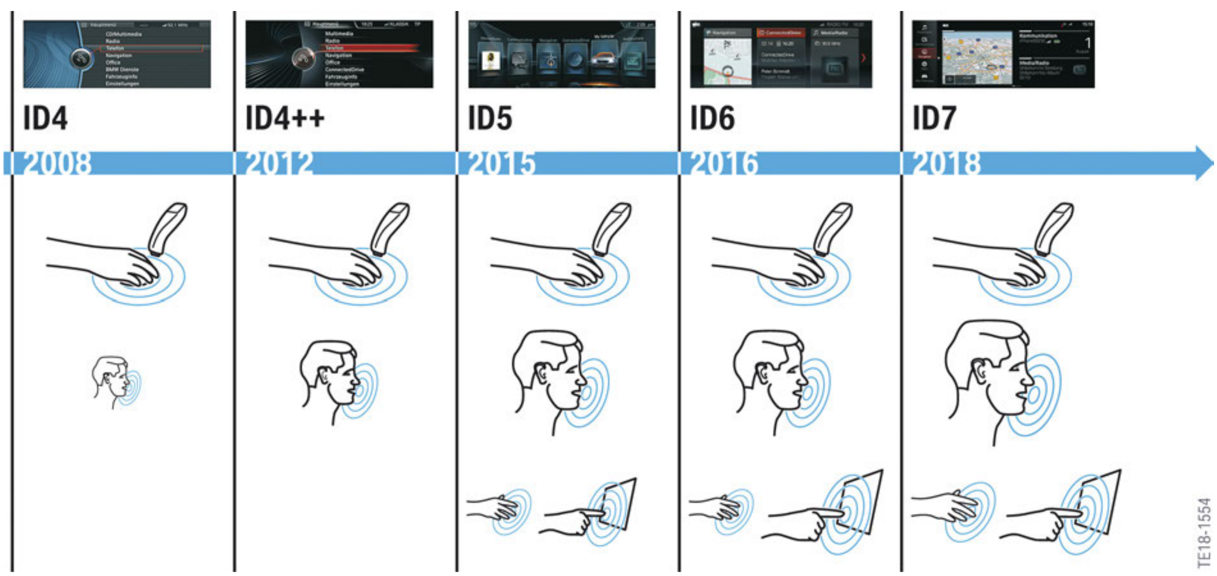
With the Life Cycle Impulse, the display and operating concept of the G12 has been retained and specifically further developed in detail. The major change is the introduction of the 7th generation BMW iDrive (ID7).

As the fundamental display elements and their operation are already known from the G12, only the changes are examined here. Additional information can be found in "ST1855 Displays and Controls 2018".

8.1. Operating elements

8.1.1. Overview

The latest generations of the operating elements are increasingly designed around several operating options. The key operating options for the ID7 include the touch control of the Central Information Display (CID), the option of a natural voice input and gesture control. The controller with touch operation is also the central operating element in vehicles with ID7.



Overview of operation of the previous generations

8.1.2. Multifunction steering wheel

For years the multifunction steering wheel has been offering a comfortable way of operating some systems. New, additional buttons and simplified operator prompting were added to the function keys on both sides.

Multimedia

Multimedia buttons on multifunction steering wheel

Index	Explanation
1	Reduce volume
2	Knurled wheel, list selection
3	LED display for driver assistance systems
4	Increase volume
5	Change station/track, long press: Fast forward the music track
6	Voice control system
7	Open entertainment lists
8	Telephone
9	Change station/track, long press: Fast reverse the music track

Further information on the operation of the multimedia systems via the multifunction steering wheel can be found in "ST1855 Displays and Controls 2018".

Driver assistance systems

Buttons for driver assistance systems on multifunction steering wheel

Index	Explanation
1	Rocker button for changing the set speed
2	LED display
3	Speed limit ON/OFF
4	Increase distance to the vehicle ahead
5	Button for selecting the assistance system Possible selection: Only ACC Stop&Go with active lane keeping assistant incl. steering & traffic jam assistant
6	Button for activating or deactivating the assistance system selected via the mode button (Assist button)
7	Cruise Control: Save speed
8	Reduce distance to vehicle ahead
9	Button for calling up a saved set speed/temporarily switching off the cruise control

Information about operating the driver assistance systems and the multifunction steering wheel can be found in "ST1831 G05 Driver Assistance Systems" and " ST1855 Displays and Controls 2018".

Steering wheel heating

G12 LCI steering wheel heating

Index	Explanation
1	Button, steering wheel heating

8.1.3. Gesture control

The current gesture control is integrated into the G12 LCI and more detailed information on the individual gestures can be found in the "ST1855 Displays and Controls 2018".

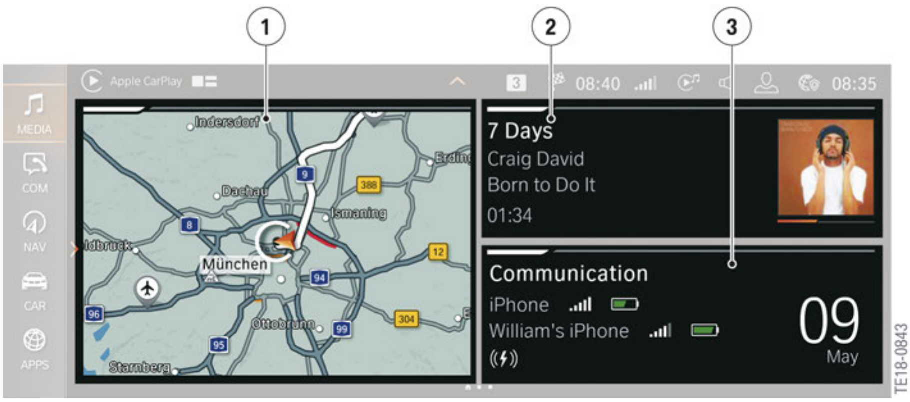
8.1.4. BMW display key

The BMW display key can be charged in the wireless charging station. This is now located in the center console (see 2.2. Interior equipment).

8.2. 7th generation iDrive (ID7)

ID7 offers a completely new display and operating structure. The tiles as they were used for ID5 or ID6 are no longer present with ID7. The displays in the main menu can be customized even more by the driver.

Up to 4 contents can be displayed on the main page. There is also the option, however, to display one content bigger, spread across half a page.



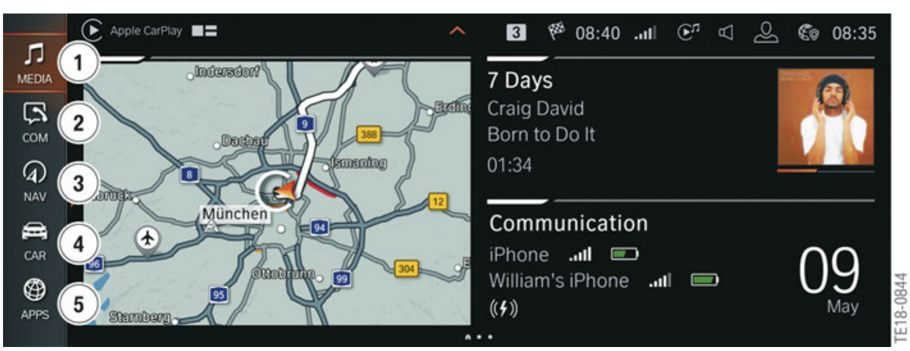
Main page of Central Information Display

Index	Explanation
1	Navigation widget
2	Radio/Media widget
3	Communication widget

8.2.1. Main menu bar

The individual menus are displayed in the toolbar in the left area of the main menu.

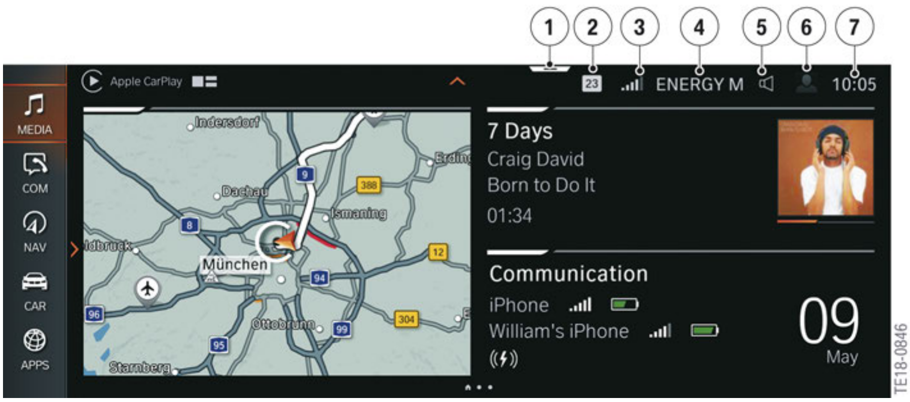
There is a total of 5 menus in the toolbar.



Menus on Central Information Display

Index	Explanation
1	Media
2	Communication
3	Navigation
4	My Vehicle
5	Apps

There is direct access to some menus. However, this can only be effected with a touch input on the Central Information Display (CID). In the following example, you see which menus can be accessed directly and what settings can be configured there:

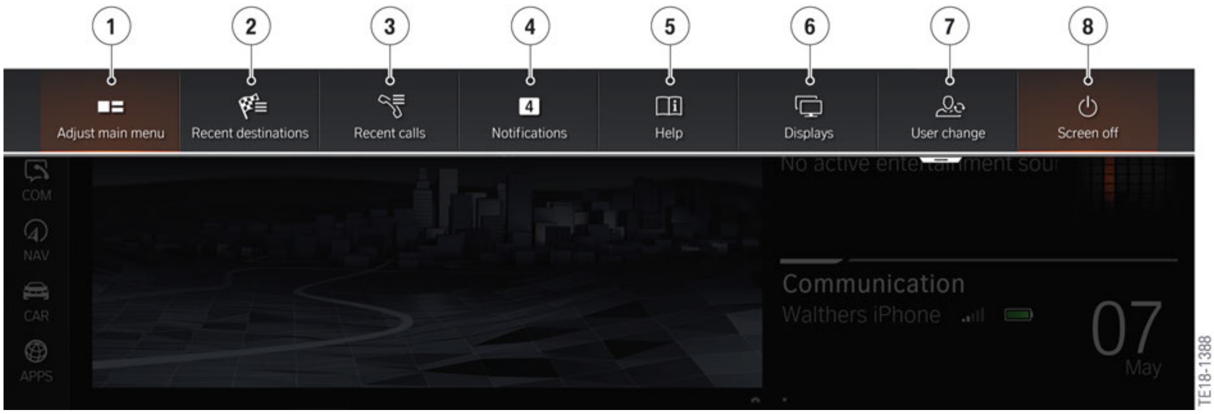


Displays on Central Information Display

Index	Display	Direct opening
1	Display bar	A description of the display bar will be provided separately
2	Messages	Read current pending or unread messages
3	Signal strength of mobile phone	Communication setting
4	Entertainment source	Media setting
5	Volume control	Sound ON/OFF
6	Profile picture	Driver profiles
7	Time	Time and date setting

8.2.2. Display bar

If the display bar is dragged down by touch, a window is opened via which certain menus or settings can be called up. The content is predefined and cannot be personalized. However, the content may vary depending on which functions are currently carried out in the vehicle. For instance, with active route guidance the point "Recent destinations" is no longer displayed.



Display bar

Index	Explanation
1	Configuration of main menu
2	Recent destinations
3	Last calls
4	Current or unread messages
5	Help
6	Display settings
7	Select driver profiles
8	Display OFF

For ID7 there is no longer a favorites view (last 20 selected menus).

Further information on ID7, operation and the sub-menus can be found in "ST1855 Displays and Controls 2018".

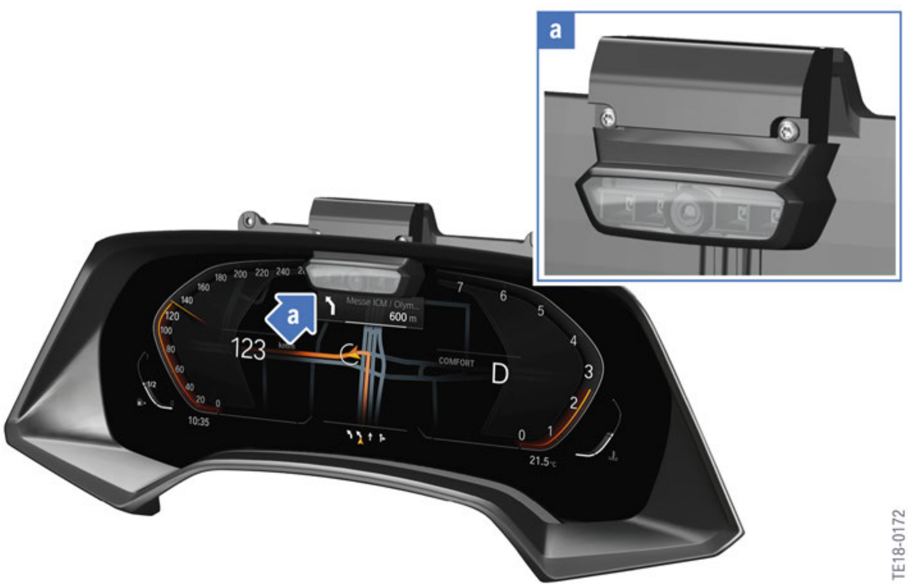
8.3. Display elements

8.3.1. Instrument panel

Driver Camera System

Depending on the equipment, a camera is installed in the upper area of the instrument cluster. The Driver Camera System (DCS) is connected directly to the instrument cluster via an Ethernet cable. The DCS monitors how open the driver’s eyes are and the alignment of the head using the nose. This information is required for certain driver assistance systems.

More information about the Driver Camera System can be found in "ST1858 Driver Assistance Systems 2018".



Driver Camera System

8.3.2. Central Information Display with touch function

The 12.3” Central Information Display (CID) now displays content with a resolution of 1920 x 720 pixels.